

Yes, process next frame

## Skip block

FrameDiff (Foreground Mask)



Generate ROI

No

empty list?

List of blocks  
at risk of fire  
or smoke.

**TINY model**  
Fire/smoke  
classifier  
for blocks

Yes

have active  
motion block?

No

Yes

SKIP\_Frame = YES

active motion  
block batch

RETURN:  
Skip\_Frame,  
ROI

Skip\_frame?

No, do infer  
**BIG model inference**

Crop input  
frame  
using ROI

*inferred using*

**BIG model**  
(complex  
model)

Video frames

